

IDS Pulse — Master System Blueprint & Architecture Guide

Complete functional breakdown and connected architectural blueprint uniting the Web Operations Suite (Command Center, Operations Hub & Client Portal) and the Mobile Quality App (Plant Floor Client).

Organization: Integrity Driven Solutions Inc. (IDS) **Platforms:** Desktop Web + Android Native APK

Production URL: <https://proud-lavoisier.vercel.app> **Standard:** Enterprise Production Grade

01 MASTER CONNECTED SYSTEM ARCHITECTURE

System Overview: IDS Pulse bridges physical plant-floor quality containment with digital executive command. The **Web Operations Suite** and the **Mobile Quality App** share an authoritative, bidirectional data synchronization bridge.

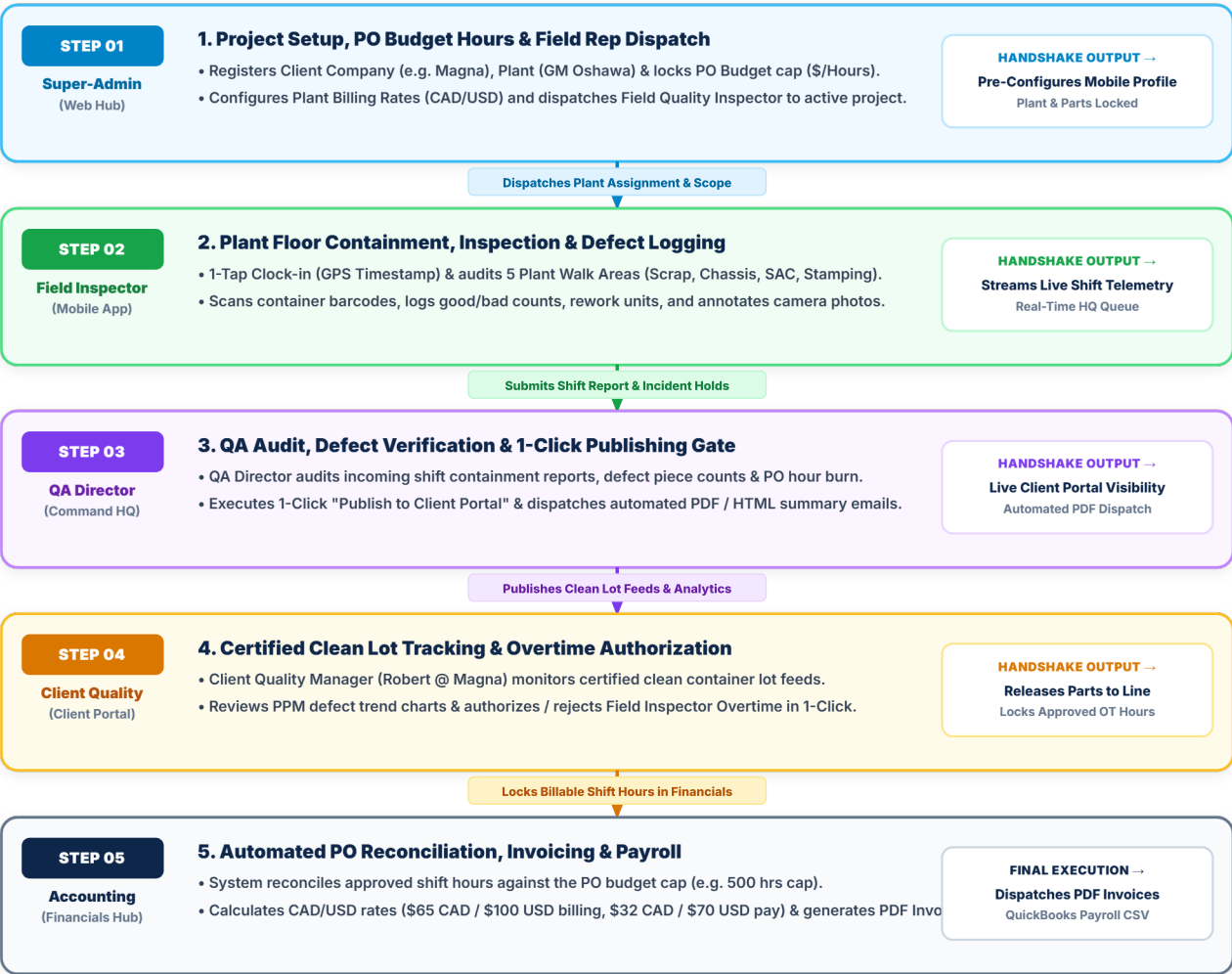
IDS PULSE PLATFORM ECOSYSTEM: CONNECTED HUB & SPOKE NETWORK BLUEPRINT



END-TO-END OPERATIONAL LIFECYCLE: 5-STEP MASTER PROCESS

FLOWCHART

Sequential data and operational handshake across all 4 User Roles from initial Project Onboarding to Final Invoicing & Payroll.



A. WEB OPERATIONS SUITE FUNCTIONS

Function ID & Name	Target Role	Core Capability & Business Purpose	Data Pipeline Connection
Universal Client Onboarding	Super-Admin	Registers Tier-1 Client Company, Assembly Plant Location, Purchase Order, and allocated Budget Hours cap in 1 modal.	Creates records in suppliers, plants, and projects collections.
User Accounts Provisioning	Super-Admin	Provisions authentication credentials for Field Inspectors, Client Quality Managers, and Staff with role presets.	Writes to users table; syncs RBAC session keys upon login.
Field Rep Dispatch Engine	Super-Admin	Dispatches Field Inspector to active plant project; locks in agreed hourly billing rate and field inspector pay rate.	Updates projects.rep_id and writes to authoritative rates table.
Master Operations Matrix	Admin / QA Dir	Central connected table linking Client, Plant, Currency (CAD/USD), Scope, PO Cap, Rep, Rates, with 1-click actions.	Queries unified data joins across clients, plants, projects, and rates.
Live Command Center Feed	QA Director	Real-time operations feed monitoring active inspector clock-ins, inspected piece counts, and open containment alerts.	Subscribes to live shiftReports and incidents event stream.
1-Click Publishing Gate	QA Director	Audit mechanism allowing QA Directors to inspect incoming draft shift reports and publish them directly to clients.	Transitions shiftReports.status from Submitted to published.
Client Executive Portal	Customer Mgr	Dedicated view for Tier-1 quality managers showing live certified clean lot tracking and PPM defect trends.	Strictly filtered by supplier_id; cost and pay rates permanently scrubbed.
Client Overtime Approval	Customer Mgr	Interactive drawer enabling customer managers to approve or decline field inspector overtime requests with notes.	Updates overtimeRequests.status to Approved / Declined.
Unified Rate Resolver	Accounting / Admin	3-Tier rate engine resolving billing and pay rates based on location (US = USD, Canada = CAD).	Executes resolveRateValue() fallback resolver across all tables.
PO Budget Burn Rate Meter	Accounting / Client	Visual progress bar tracking consumed labor hours against client PO caps with green/amber/red threshold alerts.	Calculates $\sum(\text{hours})$ from approved shiftReports against projects.allocated_hours.
Batch Invoicing & Payroll	Accounting	Generates canonical branded PDF billing invoices, QuickBooks CSV timesheets, and multi-tab payroll workbooks.	Draws directly from approved timeEntries using generateInvoicePdf.js and ExcelJS.

B. MOBILE QUALITY APP FUNCTIONS

Function ID & Name	Target Role	Core Capability & Business Purpose	Data Pipeline Connection
1-Tap Shift Clock-In	Field Inspector	Single-tap shift activation with pre-selected plant location defaults and live running elapsed hours timer.	Initializes active shift session in local storage and cloud database.
4-Step Incident Wizard	Field Inspector	Structured defect capture workflow: Step 1 Details → Step 2 Defect Quantities → Step 3 Evidence → Step 4 Routing.	Generates atomic payload for incidents and shiftReports.
Visual Evidence Canvas	Field Inspector	Integrated camera capture with canvas annotation markup tool (draw defect circles, arrows, callouts).	Encodes annotated images as Base64/Blob and attaches to incident record.
Barcode Container Scanner	Field Inspector	Rapid optical recognition of automotive container barcode labels to auto-populate part numbers.	Injects scanned string directly into part_number input field.
Defect Rework & Scrap Counter	Field Inspector	Fast counters for logging reworked units, rework labor hours, defect categories, and scrap quantities.	Persists entries in reworkLogs and updates shift summary.
3-Way Contact Routing	Field Inspector	Dynamic resolution of authorized Client Quality Contacts for selected assembly plant to receive report dispatches.	Merges client contacts table, supplier contacts, and user records.
Offline Outbox Queue	Field Inspector	Ensures uninterrupted shift logging in cellular dead-zones with automated background synchronization upon reconnect.	Managed by nativeStorageService.js and backgroundSyncWorker.js.
Overtime Request Launcher	Field Inspector	Allows inspectors on plant floors to request overtime hours with business reason directly to customer managers.	Dispatches notification and mounts pending item in Client Portal drawer.

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HOW WEB DASHBOARD & MOBILE APP CONNECT

The integration between the Web Operations Suite and Mobile Quality App follows an automated 5-step transaction lifecycle:

- STEP 1

Onboarding & Rep Assignment (Dashboard → Mobile App)
Admin onboards Client, Plant, PO Hours, and assigns the Field Rep in the **Operations Hub**. The Mobile App automatically syncs and pre-populates the inspector's plant assignment and authorized part numbers upon login.
- STEP 2

Real-Time Shift Execution (Mobile App → Dashboard)
Inspector clocks in on mobile and logs inspected parts, defect counts, rework hours, and photos. The **Live Command Center** on the dashboard updates in real time to show active plant coverage and pieces inspected.

STEP
3

QA Review & 1-Click Publishing (Mobile App → Admin → Client)

Inspector submits end-of-shift containment report. QA Director (Donna) reviews the draft in the Command Center and clicks **Publish to Client Portal**, instantly granting visibility to the automotive customer.

STEP
4

Client Visibility & Overtime Sign-off (Client Portal ↔ Mobile App)

Client Quality Manager logs into the **Client Portal**, reviews certified clean lot numbers, and approves inspector overtime requests in the approval drawer. The approval status reflects back to the inspector's timesheet.

STEP
5

Automated Financial Reconciliation (Dashboard Accounting)

Approved shift hours are reconciled against the PO budget cap. The dashboard calculates billing amounts (\$65/hr CAD or \$100/hr USD) and payroll (\$32/hr CAD or \$70/hr USD), generating single/batch PDF invoices and QuickBooks CSVs.

04 PERMISSIONS & ACCESS CONTROL MATRIX

SYSTEM CAPABILITY	SUPER-ADMIN	QA DIRECTOR	FIELD REP	CLIENT MANAGER	ACCOUNTING
Operations Hub & Setup Matrix	FULL (CRUD)	FULL (CRUD)	NO ACCESS	NO ACCESS	READ ONLY
User Provisioning & Passwords	FULL (CRUD)	FULL (CRUD)	NO ACCESS	NO ACCESS	NO ACCESS
Incident Review & Publishing Gate	PUBLISH	PUBLISH	SUBMIT ONLY	VIEW APPROVED	VIEW ONLY
Mobile Shift Clock-In & Logging	SIMULATED	SIMULATED	NATIVE ACCESS	NO ACCESS	NO ACCESS
Client Quality Portal Feed	AUDIT MODE	AUDIT MODE	NO ACCESS	PRIMARY VIEW	NO ACCESS
Billing Invoices & Payroll Export	FULL EXPORT	FULL EXPORT	NO ACCESS	INVOICE ONLY	FULL EXPORT
Field Inspector Pay Rate Data	UNRESTRICTED	UNRESTRICTED	OWN RATE	STRICT BLOCK	UNRESTRICTED

05 TECHNICAL INFRASTRUCTURE & QUALITY STANDARDS

FRONTEND & MOBILE PLATFORM

CLIENT LAYER

- **Web Operations Suite:** Modular responsive UI with high-contrast light theme tokens.
- **Mobile Quality Client:** Native mobile interface with glove-friendly touch controls.
- **Vector Graphics:** Clean SVG iconography and zero emoji enforcement.
- **Document Engines:** High-resolution PDF generation and multi-tab Excel workbooks.

BACKEND & CLOUD INFRASTRUCTURE

CLOUD LAYER

- **Database Architecture:** Relational PostgreSQL cloud storage with automated schema validation.
- **Offline Resilience:** Local transaction queuing with automated background synchronization.
- **Security & Privacy:** Strict multi-tenant client isolation and role-based permissions.
- **Deployment:** Global high-availability edge infrastructure with instant cache invalidation.